Skungrella (Lane Elien)

Getting Started with Energia IDE and the RSLK Library

This tutorial will guide you through setting up Energia IDE, installing the RSLK Library, configuring the MSP432P401R board, and running a basic blink program on the TI RSLK MAX robot.

Prerequisites

- Windows/macOS/Linux computer
- Texas Instruments (TI) MSP432P401R microcontroller
- TI RSLK MAX kit
- Micro-USB cable

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1. Download and Install Energia IDE

Energia is an open-source electronics prototyping platform based on the Arduino framework.

1. Download Energia IDE from the official website: Energia Download.
2. Unzip the downloaded file.
3. Move the extracted folder to the **Documents** directory.
4. Open the Energia IDE by double-clicking on the `energia` executable/icon.

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2. Download and Install the RSLK Library

The RSLK Library provides essential functions for controlling the TI RSLK MAX robot.

1. Download the RSLK Library from GitHub: Energia RSLK Library.
2. Save the file in the energia folder
3. Open **Energia IDE**.
4. Navigate to **Sketch** > **Include Library** > **Add .Zip Library.**
5. Use the Energia IDE to install the library

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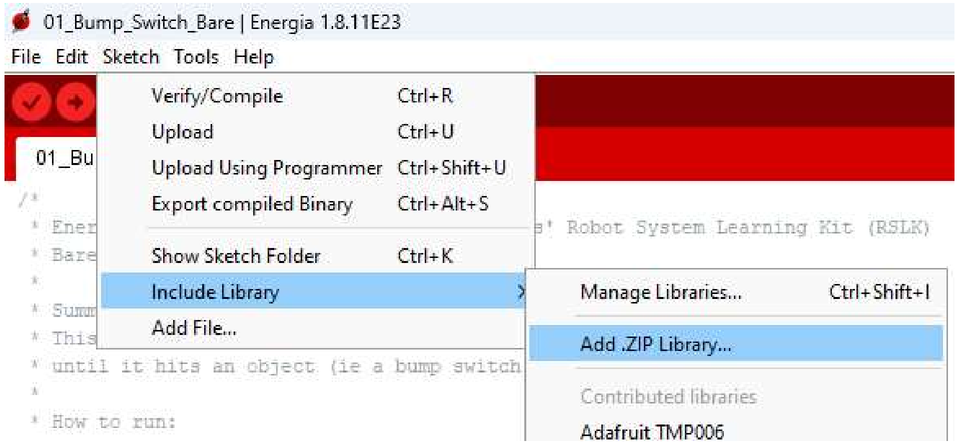

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3. Install the MSP432P401R Board Package

To program the MSP432P401R, you must install the board package in Energia.

1. Open **Energia IDE**.
2. Navigate to **Tools** > **Board** > **Boards Manager**.
3. Search for `MSP432` in the Boards Manager.
4. Select **Energia MSP432 EMT RED boards** by `Energia` and click **Install**.
5. Wait for the installation to complete.

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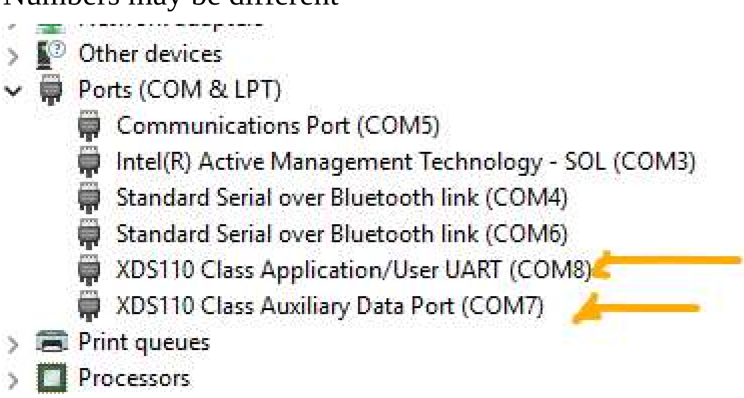
4. Connect the MSP432P401R to Your Computer

1. Use a **Micro-USB cable** to connect the MSP432P401R board to your computer.
2. Open **Device Manager** (Windows) or **System Information** (macOS) to check if the board is detected. When the board is plugged in two extra devices will appear under Ports. Your Port Numbers may be different



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5. Upload the Blink Example

The blink example tests if your setup is working by blinking the red LED on the MSP432P401R board.

1. Open Energia IDE.
2. Navigate to **File** > **Examples** > **Basics** > **Blink**.
3. Select the correct board:
    - Go to **Tools** > **Board** > **MSP-EXP432P401R (Red) LaunchPad**.
4. Select the correct port:
    - Go to **Tools** > **Port** and choose the detected port.
5. Click the **Upload** (right arrow) button to compile and upload the program.
6. If successful, the **red LED** on the board should start blinking.

Troubleshooting

- If the board is not detected, try a different USB cable.
- If the upload fails, check that the correct board and port are selected in Energia.
- Restart Energia IDE and reconnect the board if issues persist.

Next Steps

- Explore more RSLK example programs in Energia.
- Learn about motor control and sensor integration.
- Build and program custom robotic applications with the RSLK MAX.

Happy coding! 🚀